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ELECTRONIC HI-HAT CYMBAL

Abstract

An electronic hi-hat system having a movable upper housing separated from a stationary lower housing by a spring, with the upper housing having a simulated cymbal attached thereto. The upper housing and simulated cymbal connect through a central vertical rod to a lower foot pedal, with the entire structure supported by a stand on which the lower housing mounts. The lower housing has a Hall effect sensor mounted thereon, and the upper housing has a permanent magnet aligned with and arranged to reciprocate vertically alongside the Hall effect sensor. The position of the magnet on the movable upper housing, and thus the simulated cymbal, is detected by the Hall effect sensor which combines with signals generated by a sound pickup attached to the simulated cymbal that senses the cymbal being struck by a drumstick, the result being transmitted, processed, and amplified into simulated high-hat cymbal sounds

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Background/Summary

RELATED APPLICATION INFORMATION [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/887,796, filed Sep. 17, 2024, which claims priority under 35 U.S.C § 119(e) to provisional application No. 63/563,633, filed on Mar. 11, 2024.

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BACKGROUND

Field

[0003] This disclosure relates to electronic hi-hat cymbals and, more particularly, to devices for expanding and improving the range of sounds generated by an electronic hi-hat cymbal kit.

Description of the Related Art

[0004] Since the 1920s, a traditional acoustic drum set typically has a pair of metal (e.g., bronze) cymbals collectively called the hi hat. These two cymbals are suspended on a hi hat stand facing in a concave mirror image from each other. The bottom cymbal of a hi hat stays stationary, while the top cymbal can be moved up and down using a foot pedal. Modulating the force and timing of the step changes the impact of the two cymbals and thus the sound generated when the upper cymbal is struck. Depending on how hard a hi-hat is struck and whether it is “open” (i.e., pedal not pressed, so the two cymbals are not closed together), a hi-hat can produce a range of dynamics, from very quiet “chck” sounds, done with merely gently pressing the pedal to very loud crashes (e.g., striking fully open hats hard with sticks).

[0005] Electronic hi-hat cymbals are sold which don't rely on contact between acoustic bronze cymbals. A 2-piece electronic hi-hat consists of two separate components—the top cymbal and the bottom cymbal—often both equipped with sensors. These sensors detect the drummer's movements and translate them into electronic signals, which are then transduced and amplified, producing a realistic and dynamic sound.

[0006] Despite numerous examples, the electronic hi-hat devices in the art and on the market fall short in terms of quality of sounds produced.

SUMMARY OF THE INVENTION

[0007] The electronic hi-hat system disclosed herein is an assembly of two relatively moving parts coupled with electronics. A movable upper housing is separated from a stationary lower housing by a spring, with the upper housing having a simulated cymbal attached thereto. The upper housing and simulated cymbal are connected through a central vertical rod to a lower foot pedal, with the entire structure being supported by a stand on which the lower housing mounts. The lower housing has a Hall effect sensor mounted thereon, and the upper housing has a permanent magnet aligned with and arranged to reciprocate vertically alongside the Hall effect sensor. The mounting positions of the magnet and Hall effect sensor may be reversed, though the sensor comprises a circuit board which is easier to connect to associated electronics on the stationary housing. The position of the magnet on the movable upper housing, and thus the simulated cymbal, is detected by the Hall effect

sensor which combines with signals generated by a sound pickup attached to the lower surface of the simulated cymbal that senses the cymbal being struck by a drumstick, the result being transmitted, processed, and amplified into simulated high-hat cymbal sounds. Secondary sensors in the electronic hi-hat system may be used to further affect the combined result.

[0008] A first exemplary electronic hi-hat cymbal comprises a vertical cymbal stand, a stationary housing mounted at an upper end of the stand, and a foot pedal mounted to a lower end of the stand configured to be depressed by a user's foot and having a return spring. The foot pedal is connected to a vertical rod that extends up through a bore in the stand to an upper portion positioned above the stationary housing, and a simulated cymbal mounts to the upper portion of the rod. A movable housing mounts to the upper portion of the rod, wherein depression and release of the foot pedal vertically displaces the movable housing adjacent to the stationary housing, and the movable housing has a bottom position limited by impact between the movable housing and the stationary housing. A first one of the movable housing and stationary housing has a permanent magnet thereon with opposite poles vertically spaced apart and a second one of the movable housing and stationary housing has a Hall effect sensor thereon configured to generate an electronic signal, wherein the magnet and Hall effect sensor are positioned radially adjacent to each other with a radial gap therebetween so as to translate vertically relative to one other when the foot pedal is depressed. Finally, a return spring is positioned to bias the movable housing upward.

[0009] In the first exemplary electronic hi-hat cymbal, the radial gap may be 1 mm or less. The hi-hat cymbal may further include an electronic sound pickup mounted to an underside of the simulated cymbal with a lower layer of rigid plastic and an upper layer of an elastomer therebetween. The Hall effect sensor may be mounted in a stationary sidewall of the stationary housing, and the magnet is mounted in a movable sidewall of the movable housing radially outward from the Hall effect sensor.

[0010] In one example, the stationary housing has a tubular main body defining the stationary sidewall, and the movable housing has a tubular cup defining the movable sidewall that surrounds the tubular main body. In some cases, the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing. The assembly further may include at least one compressible buffer positioned between the flange and the bulkhead to cushion impact therebetween. The flange may comprise a spring foot formed of a compressible material which cushions the impact between the movable housing and the bulkhead. The spring foot may be formed with a relatively solid inner anvil portion with an annular diaphragm-like outer skirt connected and extending outward therefrom, the outer skirt positioned to contact the bulkhead before the anvil portion.

[0011] The hi-hat cymbal may further include at least one additional electronic sensor selected from the group consisting of a piezo-electric sensor and a force sensing resistor sensor, with the at least one additional electronic sensor mounted to a surface on the stationary housing that the movable housing impacts so as to generate an electronic signal upon impact between the movable housing and the stationary housing.

[0012] The hi-hat cymbal also may further include a processor connected to receive the electronic signals from the Hall effect sensor and, when connected to an amplifier, configured to generate sounds calibrated to correspond to the relative positions of the magnet and Hall effect sensor. If a spring foot mentioned above is used, the processor may be calibrated to generate a first closed sound when the outer skirt contacts the bulkhead and a different after touch sound when the anvil portion contacts the bulkhead. A spring may be positioned between the stationary housing and movable housing, with the processor calibrated to generate a first closed sound when the movable housing first contacts the spring and a different after touch sound when the movable housing compresses the spring against the stationary housing.

[0013] A second exemplary electronic hi-hat cymbal comprises a vertical cymbal stand, a stationary housing mounted at an upper end of the stand, a foot pedal mounted to a lower end of the stand configured to be depressed by a user's foot and having a return spring, the foot pedal being connected to a vertical rod that extends up through a bore in the stand to an upper portion positioned above the stationary housing, and a simulated cymbal mounted to the upper portion of the rod. A movable housing mounts to the rod below the simulated cymbal and at least partly radially adjacent the stationary housing, wherein depression and release of the foot pedal vertically displaces the movable housing and simulated cymbal. A first one of the stationary housing and movable housing has a permanent magnet thereon with opposite poles vertically spaced apart and a second one of the stationary housing and movable housing has a Hall effect sensor thereon configured to generate an electronic signal, with the magnet and Hall effect sensor being mounted on the respective housings so as to reciprocate vertically alongside each other across a radial gap when the foot pedal is depressed and released. A processor is connected to receive the electronic signals from the Hall effect sensor and, when connected to an amplifier, is configured to generate sounds calibrated to correspond to the relative positions of the magnet and Hall effect sensor.

[0014] In the second exemplary electronic hi-hat cymbal, the radial gap may be 1 mm or less. The hi-hat cymbal may further include an electronic sound pickup mounted to an underside of the simulated cymbal with a lower layer of rigid plastic and an upper layer of an elastomer therebetween. The Hall effect sensor may be mounted in a stationary sidewall of the stationary housing, and the magnet is mounted in a movable sidewall of the movable housing radially outward from the Hall effect sensor.

[0015] In one example, the stationary housing has a tubular main body defining the stationary sidewall, and the movable housing has a tubular cup defining the movable sidewall that surrounds the tubular main body. In some cases, the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing. The assembly further may include at least one compressible buffer positioned between the flange and the bulkhead to cushion impact therebetween. The flange may comprise a spring foot formed of a compressible material which cushions the impact between the movable housing and the bulkhead. The spring foot may be formed with a relatively solid inner anvil portion with an annular diaphragm-like outer skirt connected and extending outward therefrom, the outer skirt positioned to contact the bulkhead before the anvil portion.

[0016] The hi-hat cymbal may further include at least one additional electronic sensor selected from the group consisting of a piezo-electric sensor and a force sensing resistor sensor, with the at least one additional electronic sensor mounted to a surface on the stationary housing that the movable housing impacts so as to generate an electronic signal upon impact between the movable housing and the stationary housing.

[0017] If a spring foot mentioned above is used, the processor may be calibrated to generate a first closed sound when the outer skirt contacts the bulkhead and a different after touch sound when the anvil portion contacts the bulkhead. A spring may be positioned between the stationary housing and movable housing, with the processor calibrated to generate a first closed sound when the movable housing first contacts the spring and a different after touch sound when the movable housing compresses the spring against the stationary housing.

Description

DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a perspective view of a conventional pedal-operated hi-hat cymbal;

[0019] FIG. 2 is a vertical sectional view through an electronic hi-hat cymbal assembly of the present application;

[0020] FIGS. 3A and 3B are enlargements of the sectional view in FIG. 2 showing two different operating positions;

[0021] FIG. 4 is a perspective view of a movable element of a transducer assembly within the electronic hi-hat cymbal assembly of FIG. 2 carrying a magnet, shown inverted;

[0022] FIG. 5 is a perspective view of a stationary housing of the transducer assembly having a Hall effect sensor thereon;

[0023] FIG. 6A is a perspective view of an alternative stationary housing of the transducer assembly having a Hall effect sensor and supplemental sensors, and FIG. 6B is a sectional view through an electronic hi-hat cymbal assembly showing exemplary locations of the supplemental sensors;

[0024] FIG. 7 schematically illustrates a toroidal magnetic field generated by two poles of a permanent magnet;

[0025] FIG. 8 is a vertical sectional view through an electronic hi-hat cymbal assembly of the present application having a simulated lower cymbal and an “after-touch” feature;

[0026] FIGS. 9A-9D are enlargements of the sectional view in FIG. 8 showing four different operating positions;

[0027] FIG. 10 is an enlargement of a lower end of a movable element of the transducer assembly from FIG. 9A;

[0028] FIG. 11 is a schematic illustration of an electronic hi-hat cymbal having a transducer subassembly including a spring-initiated “after-touch” mechanism;

[0029] FIGS. 12A-12C are enlargements of FIG. 12 showing several positions of the transducer subassembly and springs;

[0030] FIG. 13 shows curves indicating both force and velocity experienced by the electronic hi-hat cymbal assembly of FIG. 8 graphed against the resulting effects generated by the system;

[0031] FIGS. 14A-14C are perspective views of a stationary housing of the transducer assembly having a position-adjustable Hall effect sensor thereon;

[0032] FIGS. 15A and 15B are top and bottom views of an electronic hi-hat cymbal of the present application; and

[0033] FIG. 16 is a bottom view of the electronic hi-hat cymbal in FIG. 15B with a lower housing removed, FIG. 16A is a sectional view thereof, and FIG. 16B is an enlargement showing several thin layers to improve the response of an electronic sound pickup.

[0034] Throughout this description, elements appearing in figures are assigned three-digit reference designators, where the most significant digit is the figure number and the two least significant digits are specific to the element. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described element having a reference designator with the same least significant digits.

DETAILED DESCRIPTION

[0035] FIG. 1 is a perspective view of a conventional pedal-operated hi-hat cymbal assembly 20. The hi-hat cymbal assembly 20 includes a pair of metallic, typically bronze, cymbals 22a, 22b mounted near the top of a vertical stand 24 supported by lower legs 26. While the lower cymbal 22a is fixed with respect to the stationary stand 24, the upper cymbal 22b is fixed with respect to a movable rod 28 that slides within the hollow stand 24. The rod 28 extends downward through the stand 24 and connects at its lower end to the front end of a foot pedal 30 whose rear end is mounted to pivot about a floor bracket 32. The foot pedal 30 incorporates a spring mechanism (not shown) which biases its front end upward so that a drummer need only push downward on the foot pedal to cause the cymbals 22a, 22b to come together and make an acoustic noise. The foot pedal 30 automatically returns to its raised position when the drummer's foot lifts up. As mentioned above, the velocity and force applied to the foot pedal 30 can be varied to modulate the acoustic sound.

[0036] FIG. 2 is a vertical sectional view through an electronic hi-hat cymbal assembly **40** of the present application. Although not shown completely, the hi-hat cymbal assembly **40** is mounted at the top of a stationary vertical stand **42** supported on the floor with legs, for example. The stand **42** is hollow and a rod **44** is arranged to move up and down within an inner throughbore. The rod **44** connects to a lower foot pedal, such as the ones shown in FIG. 1, to actuate the hi-hat cymbal assembly **40**.

[0037] The rod **44** extends upward beyond the top of the stand **42** and is fixed to an upper housing **46** to which an upper cymbal **48** mounts. The cymbal **48** is a simulated cymbal as it does not actually strike a lower cymbal to make or modify an acoustic sound. Indeed, in this embodiment there is no lower cymbal as the sound produced does not depend on the acoustic sound generated by the impact or character of contact between two physical cymbals. Typically, the cymbal **48** is made of a polymer, though any lightweight material may be utilized. Alternatively, bronze or other metallic materials may be used, with the simulated cymbal being a solid disk, perforated, or generally formed from any rigid materials and with any configuration available on the market. The simulated cymbal just does not need to strike or contact a paired cymbal or be struck with a drumstick to make a variety of sounds.

[0038] An upper nut **50** may be fastened to a top end of the rod **44** to hold the cymbal **48** against the upper housing **46**. The upper housing **46** moves up and down with the rod **44** relative to a lower housing **52**, carrying the cymbal **48** with it. The lower housing **52**, in turn, is fixed with respect to a bulkhead **54** mounted to the top of the stationary stand **42**. The bulkhead **54** has a vertical bore through which the rod **44** slidingly reciprocates. The upper housing **46** and cymbal **48** thus move up and down with respect to the bulkhead **54** on which the lower housing **52** is mounted.

[0039] The upper housing **46** and the lower housing **52** formed the main components in a transducer subassembly of the hi-hat cymbal assembly **40**, and are shown in perspective in FIGS. 4 and 5. The stationary lower housing **52** has a generally tubular main body **60** with three tabs **62** extending outward from an upper edge thereof. The bulkhead **54** widens outward from a lower end of the tubular main body **60**. Each of the tabs **62** has an outer ramp surface **64** and is mounted in a cantilevered fashion due to vertical slits **66** formed in the tubular main body **60**. The upper housing **46** is shown inverted in FIG. 4 to illustrate a horizontal flange **70** extending outward from a lower end of a generally tubular cup **72** and surrounding a cylindrical inner cavity **74** within the main body. The main body includes three vertical slots **76** extending longitudinally along a majority of its length.

[0040] The inner cavity **74** of the upper housing **46** is configured to fit downward over the tubular main body **60** of the lower housing **52** and be retained thereon. More particularly, the cavity **74** closely surrounds the tubular main body **60** of the lower housing **52**, and as the upper housing **46** is pressed downward onto the main body of the lower housing **52** a lower circular rim **78** around the inner cavity **74** contacts the outer ramp surfaces **64** of each of the tabs **62**. Downward movement of the upper housing **46** cams the cantilevered tabs **62** inward to permit the upper housing to descend down around the lower housing main body **60**. At some point, the three tabs **62** flex back outward into the vertical slots **76** in the tubular cup **72** of the upper housing **46**. In this way, the upper housing **46** is captured by the lower housing **52**, but may move up and down by virtue of the tabs **62** within the elongated slots **76**. This arrangement also prevents relative rotational movement of the upper housing **46** as it slides up and down over the lower housing **52**. The movable upper housing **46** has a down position limited by the stationary lower housing **52**. More particularly, the horizontal flange **70** comes into contact with the wider bulkhead **54** which stops further downward movement of the upper housing **46** and cymbal **48** mounted thereon.

[0041] With reference back to FIG. 2, a coil spring **80** is mounted around the rod **44** between the movable upper housing **46** and the stationary lower housing **52** to bias the upper housing upward. A soft washer or buffer **82** made of a compressive material such as rubber is desirably placed on the bulkhead **54** to cushion the impact of the flange **70** of the upper housing **46** when it displaces

downward relative to the lower housing **52** with force. The buffer **82** may be annular and extend evenly around and between the bottom face of the flange **70** and the top face of the bulkhead **54**. Alternatively, a series of buffers **82** may be distributed between the two impacting surfaces. [0042] FIG. **2** also shows a magnet **84** carried by the upper housing **46**. The magnet **84** mounts to the inside of the tubular cup **72** of the upper housing **46** so as to be exposed within the cavity **74**, as seen in FIG. **4**. The magnet **84** may be shaped as an elongated vertical rectangle, as shown. The magnet **84** is aligned with and translates up and down relative to and alongside a Hall effect sensor **86** mounted to the tubular main body **60** of the lower housing **52**, seen in FIG. **5**. Due to the concentric nature of the tubular main body **60** and tubular cup **72**, the magnet **84** lies radially outside the sensor **86** and translates alongside it with a radial gap of 1 mm or less therebetween permitting relative sliding movement. The vertical travel of the rod **44** and movable upper housing **46** may be about 18 mm which may correspond to the vertical dimension of the sensor **86**, with the magnet **84** remaining adjacent to the sensor **86** to prevent magnetic decoupling. Stated another way, the magnet **84** is circumferentially aligned with the sensor **86** and reciprocates up and down radially outside of the sensor **86** across the radial gap.

[0043] It should be understood that the structural configurations of the movable upper housing **46** in terms of a generally tubular cup **72** surrounding a cylindrical inner cavity **74** that closely and concentrically surrounds the tubular main body **60** of the lower housing **52** are exemplary only. That is, the position of the magnet **84** that slides alongside and radially outside the sensor **86** may be established with alternative relatively movable structures, and so the terms upper housing and lower housing should be seen as representative of a variety of solid objects, one movable and one stationary, or vice versa. For instance, the concentric tubular structures could be replaced with flat or curved vertical walls that are spaced closely apart and on which are mounted the respective magnet **84** and sensor **86**.

[0044] The Hall effect sensor **86** comprises a circuit board and may be obtained off-the-shelf from various vendors, such as a DRV5056 Unipolar Ratiometric Linear Hall Effect Sensor from Texas Instruments, of Dallas, TX. The DRV5056 Hall effect sensor has a detection range is in the region of 18 mm. As the magnet **84** translates alongside the Hall effect sensor **86**, the sensor generates varying electronic signals when the upper cymbal **48** is struck. By calibrating these electronic signals with an understanding of the relative positions of the movable upper housing **46** and the stationary lower housing **52**, and converting them using a processor and an amplifier (not shown), distinctive desirable hi-hat sounds can be produced. For instance, the point at which the horizontal flange **70** of the upper housing **46** comes into contact with the bulkhead **54** of the lower housing **52** corresponds to a relative position between the magnet **84** and the Hall effect sensor **86**. The mounting positions of the magnet **84** and Hall effect sensor **86** may be reversed, though the sensor comprises a circuit board which is easier to connect to associated electronics if mounted on the stationary housing.

[0045] FIGS. **3A** and **3B** are enlargements of the sectional view in FIG. **2** showing two different operating positions and the relative positioning between the magnet **84** and the Hall effect sensor **86**. The magnet **84** is arranged to have two vertically separated poles-indicated with S for South and N for North. FIG. **3B** shows the lower flange **70** of the movable upper housing **46** as it contacts the soft buffer **82** on the bulkhead **54** of the stationary lower housing **52**, which is the “closed” position as indicated by the schematic position list to the left. As with acoustic cymbals coming together, particular sounds are generated if, when and how the movable housing **46** contacts the stationary housing **52**, which is felt by the drummer's foot on the pedal, when the upper cymbal **48** is struck. Of course, the distance the movable housing **46** is raised and its velocity in striking the stationary housing **52** are also factors in the sound produced, all of which are determined by movement of the magnet **84** as detected by the Hall effect sensor **86**.

[0046] FIG. **6A** is a perspective view of an alternative stationary housing **52'** of the transducer assembly having both a Hall effect sensor **86** and a pair of supplemental sensors **90**, **92**, and FIG.

6B is a sectional view through an electronic hi-hat cymbal subassembly showing exemplary locations of the supplemental sensors. The supplemental sensors **90**, **92** provide alternative/additional methods of control of the sound output of any of the hi-hat cymbal assemblies described herein. Added sensors may be utilized to match the “mechanical feel” of the hi-hat cymbal to the sensor output, and may be useful to sense the exact point where the closed pedal hits the closed stop.

[0047] In one example, a piezo-electric sensor **90** is added to the stationary housing **52'** or anvil of the assembly. The piezo-electric sensor **90** may comprise an annular disk-shaped element with a flexible central portion supported around the perimeter. The central flexible portion, or diaphragm, is bent slightly when the movable housing **46** descends and slams into the rubber buffer **82** provided for cushioning. The central portion of the piezo-electric sensor **90** may be only 1 mm thick and needs to flex only a fraction of a millimeter to output a change in voltage. This small voltage change can then be read by the associated electronics to give an indication of exactly when and at what velocity the pedal was closed. Consequently, the piezo-electric sensor **90** provides a highly reliable and accurate signal defining when the pedal hits the “closed” position, and how fast or hard the pedal is closed.

“After Touch” Sound Effects

[0048] On acoustic cymbals, there is a particular sound when the two cymbals are closed together and struck, and then there is a different sound when the pedal is pressed with more force—a so-called “after touch” sound. The sound decay time gets shorter and the pitch goes up. The present application contemplates several ways to simulate the “after touch” sounds. In a first technique, the signal range of the Hall sensor can be set, but a bottom end of the range can be exceeded if the user presses with more force. This would be accomplished through calibrating the software so that an initial closed position of the magnet relative to the Hall sensor corresponds to a closed sound, while further pressing of the pedal moves the magnet slightly farther against a physical resistance force which generates the tightly closed or “after touch” sound effect. The sound generator might just shorten the decay and raise the pitch of the closed sound, or a different simulated sound sample altogether could be triggered. The physical resistance force, which may be a simple spring, provides tactile feedback. In a second technique, the bottom end of the signal range of the Hall sensor can be set to the farthest point of magnet travel, but pressing down hard on the pedal will be detected by a special pressure sensor. Both of these techniques will be illustrated below.

[0049] With reference again to FIGS. **6A** and **6B**, a supplemental force sensing resistor (FSR) sensor **92** may be added to the stationary housing **52'**. The FSR sensor **92** is similar to the piezo-electric sensor **90** in that it senses small changes in pressure. A typical FSR sensor **92** consists of two small mylar sheets, one of which has silver interdigits printed on it and the other a resistive ink. When the resistive ink presses against the interdigits, a circuit is formed, and the resistance of the circuit varies according to the pressure. Consequently, as the pedal of the hi-hat assembly is pressed down, a voltage is derived by the FSR sensor **92** that is proportional to the pressure. The voltage generated by the FSR sensor **92** is an accurate point at which the pedal is closed, but also provides a wide controller signal for any “after pressure” or “after touch” exerted on the pedal. That is, there may not be enough range in the output of the Hall effect sensor **86** after the pedal is closed to get a desired signal for the aftertouch sound. The FSR sensor **92** gets squeezed after the pedal is fully closed and provides additional control for this aftertouch sound.

[0050] As seen in the cross-section section of FIG. **6B**, either or both of the piezo-electric sensor **90** and FSR sensor **92** may be fitted in shallow wells on the top surface of the bulkhead **54**. Pressure to the sensors is thus transmitted through the rubber buffers **82** from the downwardly moving housing **46**. Both sensors **90**, **92** are connected via wires (not shown) to the sound control system.

Satisfactory performance of the electronic hi-hat cymbals described herein may be attained with just the Hall effect sensor **86**. However, more accuracy and expression-so-called “pro-level performance”—may be attained with the addition of one or both of the sensors **90**, **92**. That is, the

Hall effect sensor **86** works by software predicting the position of the moving housing **46** through a changing magnetic field, and adding the sensors **90**, **92** which respond to physical feedback provides additional accuracy.

[0051] As is well known in the art, and schematically illustrated in FIG. 7, the two poles of the magnet **84** generate a toroidal magnetic field having a horizontal midplane M coinciding with the junction of the two poles. That is, the magnetic field magnetic dipole moment, or strength, points in the direction of lines between the South and North poles of the magnet. Relative to the Hall effect sensor **86**, the direction of the magnetic field switches at the horizontal midplane M between the two poles, which transition can be sensed by the sensor. Moreover, the Hall effect sensor **86** is calibrated to sense the strength of the magnetic field proportional to the distance from the midplane M between the two poles, as well as the velocity and acceleration of the relative movement between the sensor and the magnet and output a voltage proportional to this movement. The resulting signal generated by the Hall effect sensor **86** is then processed using specialized software and converted into electric signals which can be amplified through speakers for the different hi hat sounds. Of course, the relative vertical position of the poles (South up or down) is reversible.

[0052] Additionally, though not shown, piezo sensor and position switches may be incorporated into the playing surface as per current designs. The number of position switches varies, and could be as few as two (bell and edge) and as many as 5 (bell, hi bow, med bow, low bow & edge).

[0053] FIG. 8 is a vertical sectional view through an electronic hi-hat cymbal assembly **80** of the present application having a simulated lower cymbal as well as an “after-touch” sound feature. Although not shown completely, much like the assembly of FIG. 2, the hi-hat cymbal assembly **100** is mounted at the top of a stationary vertical stand **102** supported on the floor with legs, for example. The stand **102** is hollow and a rod **108** is arranged to move up and down within an inner throughbore. The rod **108** connects to a lower foot pedal, such as the ones shown in FIG. 1, to actuate the hi-hat cymbal assembly **100**.

[0054] The rod **108** extends upward beyond the top of the stand **102** and is fixed to an upper housing **114** to which an upper cymbal **104** mounts. The cymbal **104** is a simulated cymbal as it does not strike a lower cymbal to make sound. Typically, the cymbal **104** is made of a polymer, though any lightweight material may be utilized. An upper nut **106** may be fastened to a top end of the rod **108** to hold the cymbal **104** against the upper housing **114**. The upper housing **114** moves up and down with the rod **108** relative to a lower housing **110**, carrying the cymbal **104** with it. The lower housing **110**, in turn, is fixed with respect to a simulated lower cymbal **112** mounted to the top of the stationary stand **102**. A coil spring **116** is mounted around the rod **108** between the movable upper housing **114** and the stationary lower housing **110** to bias the upper housing upward. The upper housing **114** and cymbal **104** thus move up and down with respect to the lower housing **110** and simulated cymbal **112**, simulating an acoustic cymbal assembly such as in FIG. 1. FIG. 8 shows a stationary rubber cushion or buffer **118** placed on a top face of the lower cymbal **112**, and a spring foot **120** at the lower end of the movable upper housing **114**, whose purposes are described below.

[0055] FIG. 8 also shows a magnet **124** carried by the upper housing **114**. The magnet **124** to a main body of the upper housing **114** so as to be exposed within an inner cavity of the housing **114**, much as with the earlier embodiment of FIG. 2. The magnet **124** is aligned with and translates up and down relative to a Hall effect sensor **126** mounted to the lower housing **110**, seen enlarged in FIG. 9A. The Hall effect sensor **126** comprises a circuit board and may be obtained off-the-shelf from various vendors, as mentioned above. As the magnet **124** translates alongside the Hall effect sensor **126**, the sensor generates varying electronic signals. By calibrating these electronic signals and converting them using a processor and an amplifier (not shown), distinctive desirable hi-hat sounds can be produced.

[0056] FIGS. 9A-9D are enlargements of the sectional view in FIG. 8 showing different operating positions. FIG. 10 is an enlargement of a lower end of the movable housing **114** of the transducer

assembly from FIG. 9A, illustrating the lower elastic annular molding or spring foot **120**. The foot **120** is a compressible material which cushions and influences the position of the movable housing **114** when it contacts the stationary rubber buffer **118** directly below it. More specifically, the spring foot **120** has a thick relatively solid inner anvil portion **121** with an annular diaphragm-like outer skirt **122** connected and extending outward therefrom. There also may be a soft annular washer **124** (FIG. 10) made of a compressive material such as rubber placed between flanges of the movable upper housing **114** and the stationary housing **114** to absorb forces and reduce associated noise when the upper cymbal **104** and upper housing **114** are released to move upward under influence of the spring **116**.

[0057] The annotation in FIG. 9A indicates that there is a first distance of travel between full open and full closed such as 16 mm, but a total second travel distance of 18 mm which takes into account travel after the spring foot **120** contacts the stationary rubber buffer **118**. There is thus a 2 mm after touch travel. These distances may be altered to create different cymbal actions, and are used as an example here to match the Texas Instruments DRV5056 Hall effect sensor mentioned above.

[0058] In FIG. 9B, the pedal is depressed lightly down to close the hi-hat and create a so-called “loose sizzle.” In this position, the outer skirt **122** of the spring foot **120** has come into contact with the buffer **118**, but the solid inner anvil portion **121** remains slightly elevated above the buffer **118**. This position and movement is calibrated into the sensor **126** to create the “loose sizzle” sound. The drummer feels the resistance from contact of the outer skirt **122** of the foot **120** with the buffer **118**, indicating to him/her the hi-hat is nearly closed.

[0059] In FIG. 9C, the pedal is depressed firmly down to close the hi-hat. In this position, the outer skirt **122** of the foot **120** flexes such that the inner anvil portion **121** comes into contact with the buffer **118**. This position and movement is calibrated into the sensor **126** to create the closed sound of the cymbal assembly **80**, and the drummer feels solid resistance from contact of the inner anvil portion **121** with the buffer **118**, indicating to him/her the hi-hat is nominally closed.

[0060] Finally, FIG. 9D shows the transducer assembly after the pedal has been depressed down hard to close the hi-hat and push the inner anvil portion **121** of the foot **120** downward into the compressible buffer **118**. This is also a closed sound, but the sensor **126** is calibrated to generate a slightly higher pitch of sound, such as one semitone up. Each of these positions results in a different cymbal sound.

[0061] FIG. 11 is a schematic illustration of an electronic hi-hat cymbal system including a spring-initiated “after-touch” mechanism. The hi-hat cymbal assembly has a transducer subassembly with a simulated cymbal **132** attached to an upper housing **134** movable with a vertical shaft or rod (not shown through a bore in a stationary lower housing **136**. The rod may be actuated by a foot pedal as is known. One of the movable upper housing **134** or stationary lower housing **136** carries a magnet (not shown), while the other carries a Hall effect sensor (also not shown). Typically, the magnet is on the movable upper housing **134**. As described herein, the magnet and Hall effect sensor are positioned to reciprocate up and down radially adjacent to one another when the rod moves the upper housing **134** relative to the lower housing **136**. Various Hi-Hat sounds are generated when the cymbal **132** is struck.

[0062] The stationary lower housing **136** extends upward from a bulkhead **138** aligned below the movable upper housing **134**. One or more springs **140** positioned on top of the bulkhead **138** are compressed when the upper housing **134** moves downward. The springs **140** may be simple coil springs, as depicted, or may be supplied by the spring foot **120** and compressible buffer **118** as seen in FIGS. 8-10. The springs **140** supply physical feedback to the drummer who can feel the extent of spring resistance and thus knows when the hi-hat is “closed” and how hard.

[0063] FIGS. 12A-12C are enlargements of FIG. 12 showing several positions of the transducer subassembly **130** and springs **140**. In FIG. 12A, the movable upper housing **134** is above the springs **140** indicating that the simulated cymbal **132** is loose and free of contact with any lower

cymbal. FIG. 12B shows a moment right after contact of the underside of the movable upper housing **134** with the springs **140**, which is felt by the drummer and indicates a closed position. The programming of the software connected to the Hall effect sensor is calibrated to produce sounds associated with a closed pair of cymbals. Finally, FIG. 12C shows the movable upper housing **134** dropped down farther to fully compress the springs **140**, which happens when the drummer stamps on the foot pedal and corresponds to a tightly closed cymbal position. This “after-touch” position is also calibrated into the software receiving signals from the Hall effect sensor to generate different sounds, typically a shorter decay time and a higher pitch.

[0064] This schematic representation illustrates how the programming of the sound generation software may be customized to fit the physical feedback character of the electronic hi-hat cymbal system. Nuances such as dual-springs with different spring constants may be used, for example. Additionally, once the upper housing **134** contacts the springs **140**, several steps to fully closed may be included to raise of the pitch of the sample when the pedal is pressed harder. There may not be a fixed sharpening of the pitch, but a variable amount dependent upon the pressure applied to the pedal. For instance, the software could be calibrated to raise the pitch by about a tone between spring contact and fully closed, in a minimum of 8 steps. It should also be noted that the FSR sensor **92** seen in FIG. 6A works in a similar manner. That is, because the FSR sensor **92** senses small changes in pressure and generate a voltage that is proportional to the pressure, it provides a wide controller signal for any “after pressure” or “aftertouch” exerted on the pedal.

[0065] FIG. 13 show curves indicating both force and velocity experienced by the electronic hi-hat cymbal of FIG. 8 graphed against the resulting effects generated by the system. The distances range from fully open to the left on the X-axis to closed and then farther in the “aftertouch” range, which is after the inner thicker portion of the foot **120** contacts the buffer **118**. The solid line curve V indicates the output from the Hall effect sensor **126**. The two force curves F1 and F2 correspond to the mechanical feedback exerted on the pedal at different stages of foot depression. An initial force curve F1 is felt during travel from the open to the “loose sizzle” commencement when the outer skirt **122** of the foot **120** has come into contact with the buffer **118**. Subsequently, the rate of force feedback increases in the second force curve F2 through the sizzle zone and into the aftertouch zone. These zones correspond to flexing of the elastomeric foot **120** and then impact of the inner anvil portion **121** of the foot **120** with the buffer **118** and deformation thereof. Various nodes or points are indicated on the curves to denote the various transitions, which points are calibrated into the software receiving the Hall effect signals to determine the desired sound created.

[0066] FIGS. 14A-14C are perspective views of a stationary housing of the transducer assembly having a position-adjustable Hall effect sensor **86'** thereon. As mentioned above, the Hall effect sensors described herein produce signals which are then processed using software which interprets the relative position of the moving members of the hi-hat assembly. The software can of course be calibrated at the manufacturing stage for the particular physical arrangement. Additionally, the software may possess the ability for the user to adjust the calibration settings for different sound effects. However, it may also be useful to supplement this virtual calibration with a physical calibration means. There is, the “closing point” of the hi-hat cymbal is critical for the correct feel. It is important for the software to recognize the exact point where the mechanical movement of the pedal hits the rubber covered anvil and produce a closing sample sound. Adding a mechanical way of adjusting the vertical position of either the magnet or the Hall effect sensor to adjust the calibration of the closing point is thus contemplated.

[0067] In FIGS. 14A-14C the position-adjustable Hall effect sensor **86'** comprises a small circuit board **150** as before mounted on a fixed panel **152** within a vertical slot **154** of the stationary housing. A threaded rod **156** is arranged to displace the circuit board **150** up and down relative to the panel **152**, preferably by at least a few millimeters. For example, the circuit board **150** and panel **152** may have cooperating vertical rails (not shown). The threaded rod **156** may pass through a similarly threaded bore (not shown) in the circuit board **150** so that rotation of the rod causes

vertical movement of the circuit board. The threaded rod **156** may pass downward through the bulkhead **54** to a lower thumbscrew or other actuator (not shown) accessible to the user. FIG. **14B** shows the circuit board **150** being displaced downward, while FIG. **14C** shows the circuit board displaced upward. Slight adjustments may be made in conjunction with the software calibration to control the desired sound effects.

[0068] FIGS. **15A** and **15B** are top and bottom views of an electronic hi-hat cymbal **170** of the present application. The cymbal **170** comprises a generally disk-like body (often made of plastic) with an off-center throughhole **172** for receiving a rod (not shown, such as the rod **44** shown in FIG. **2**) connected to a lower foot pedal which moves the cymbal **170** up and down and actuates the electronics of a hi-hat cymbal assembly as described herein. The cymbal **170** may have an upper surface **174** of various configurations, such as concentric ribs of rubber as shown which helps dampen the impact of a drumstick and enhance the resulting sound produced by the electronics. [0069] FIG. **15B** shows the lower surface **176** and a non-circular orifice **178** which receives a similar-shaped bushing on the rod to prevent rotation of the cymbal **170**. A rigid housing **180** is provided to cover a sound pickup attached to the lower surface **176** as will be explained, with an electronic connector jack **182** shown on one side of the housing. The rigid housing **180** also covers over the non-circular orifice **178**.

[0070] FIG. **16** is a bottom view of the electronic hi-hat cymbal **170** with the lower housing **180** removed to expose an electronic sound pickup **182**. The sound pickup **182** attaches firmly to the lower surface of the simulated cymbal **170** and senses and produces signals in conjunction with the input from the Hall effect sensor and potentially secondary sensors when the cymbal is struck from above by a drumstick.

[0071] The sound pickup **182** connects to an adjacent circuit board **184** and is mounted to the bottom of a rigid sound damping plate **186**. FIGS. **16A** is a sectional view thereof, and FIG. **16B** is an enlargement showing several thin layers to improve the response of the sound pickup **182**. Namely, the pickup **182** mounts on the rigid sound damping plate **186** which in turn mounts to the lower surface **176** of the cymbal **170** with a further elastomeric sound damping sheet **188** interposed therebetween. The rubber damping sheet **188** muffles the sound and protects the sound pickup **182**.

[0072] The rigid sound damping plate **186** may be plastic such as PVC and the combination with the elastomeric sound damping sheet **188** reduces the tendency of the sound pickup **182** to be affected by the striking of the cymbal by a drumstick directly above the pickup. The sound pickup **182** may be a piezoelectric microphone, which is often used in electronic cymbals, and the combined layers of PVC **186** and rubber **188** between the pickup **182** and cymbal **176** reduces any “hot spots” that occur when someone plays directly above the piezo, versus playing slightly to either side. The drummer then does not hear a dramatic volume change when they play a few millimeters to one side or another, but instead a consistent volume like they would hear on an acoustic, metal cymbal.

Closing Comments

[0073] Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

[0074] As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed

transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

Claims

1. An electronic hi-hat cymbal, comprising: a vertical cymbal stand; a stationary housing mounted at an upper end of the stand; a foot pedal mounted to a lower end of the stand configured to be depressed by a user's foot and having a return spring, the foot pedal being connected to a vertical rod that extends up through a bore in the stand to an upper portion positioned above the stationary housing; a simulated cymbal mounted to the upper portion of the rod; a movable housing mounted to the upper portion of the rod, wherein depression and release of the foot pedal vertically displaces the movable housing adjacent to the stationary housing, and the movable housing has a bottom position limited by impact between the movable housing and the stationary housing; wherein a first one of the movable housing and stationary housing has a permanent magnet thereon with opposite poles vertically spaced apart and a second one of the movable housing and stationary housing has a Hall effect sensor thereon configured to generate an electronic signal, wherein the magnet and Hall effect sensor are positioned radially adjacent to each other with a radial gap therebetween so as to translate vertically relative to one other when the foot pedal is depressed; and a return spring positioned to bias the movable housing upward.
2. The hi-hat cymbal of claim 1, wherein the radial gap is 1 mm or less.
3. The hi-hat cymbal of claim 1, further including an electronic sound pickup mounted to an underside of the simulated cymbal with a lower layer of rigid plastic and an upper layer of an elastomer therebetween.
4. The hi-hat cymbal of claim 1, wherein the Hall effect sensor is mounted in a stationary sidewall of the stationary housing, and the magnet is mounted in a movable sidewall of the movable housing radially outward from the Hall effect sensor.
5. The hi-hat cymbal of claim 4, wherein the stationary housing has a tubular main body defining the stationary sidewall, and the movable housing has a tubular cup defining the movable sidewall that surrounds the tubular main body.
6. The hi-hat cymbal of claim 5, wherein the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing, and further including at least one compressible buffer positioned between the flange and the bulkhead to cushion impact therebetween.
7. The hi-hat cymbal of claim 5, wherein the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing, and wherein the flange comprises a spring foot formed of a compressible material which cushions the impact between the movable housing and the bulkhead.
8. The hi-hat cymbal of claim 7, wherein the spring foot is formed with a relatively solid inner anvil portion with an annular diaphragm-like outer skirt connected and extending outward therefrom, the outer skirt positioned to contact the bulkhead before the anvil portion.
9. The hi-hat cymbal of claim 1, further including at least one additional electronic sensor selected

from the group consisting of a piezo-electric sensor and a force sensing resistor sensor, the at least one additional electronic sensor being mounted to a surface on the stationary housing that the movable housing impacts so as to generate an electronic signal upon impact between the movable housing and the stationary housing.

10. The hi-hat cymbal of claim 1, further including a processor connected to receive the electronic signals from the Hall effect sensor and, when connected to an amplifier, configured to generate sounds calibrated to correspond to the relative positions of the magnet and Hall effect sensor.

11. An electronic hi-hat cymbal, comprising: a vertical cymbal stand; a stationary housing mounted at an upper end of the stand; a foot pedal mounted to a lower end of the stand configured to be depressed by a user's foot and having a return spring, the foot pedal being connected to a vertical rod that extends up through a bore in the stand to an upper portion positioned above the stationary housing; a simulated cymbal mounted to the upper portion of the rod; a movable housing mounted to the rod below the simulated cymbal and at least partly radially adjacent the stationary housing, wherein depression and release of the foot pedal vertically displaces the movable housing and simulated cymbal; wherein a first one of the stationary housing and movable housing has a permanent magnet thereon with opposite poles vertically spaced apart and a second one of the stationary housing and movable housing has a Hall effect sensor thereon configured to generate an electronic signal, with the magnet and Hall effect sensor being mounted on the respective housings so as to reciprocate vertically alongside each other across a radial gap when the foot pedal is depressed and released; and a processor connected to receive the electronic signals from the Hall effect sensor and, when connected to an amplifier, configured to generate sounds calibrated to correspond to the relative positions of the magnet and Hall effect sensor.

12. The hi-hat cymbal of claim 11, wherein the radial gap is 1 mm or less.

13. The hi-hat cymbal of claim 11, further including an electronic sound pickup mounted to an underside of the simulated cymbal with a lower layer of rigid plastic and an upper layer of an elastomer therebetween.

14. The hi-hat cymbal of claim 11, wherein the Hall effect sensor is mounted in a stationary sidewall of the stationary housing, and the magnet is mounted in a movable sidewall of the movable housing radially outward from the Hall effect sensor.

15. The hi-hat cymbal of claim 14, wherein the stationary housing has a tubular main body defining the stationary sidewall, and the movable housing has a tubular cup defining the movable sidewall that surrounds the tubular main body.

16. The hi-hat cymbal of claim 15, wherein the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing, and further including at least one compressible buffer positioned between the flange and the bulkhead to cushion impact therebetween.

17. The hi-hat cymbal of claim 16, wherein the stationary housing has a lower bulkhead at a bottom end of and extending radially outward from the tubular cup, and the movable housing has a lower flange at a bottom end of and extending radially outward from the tubular cup, wherein the flange contacts the bulkhead to define a down position of the movable housing, and wherein the flange comprises a spring foot formed of a compressible material which cushions the impact between the movable housing and the bulkhead.

18. The hi-hat cymbal of claim 17, wherein the spring foot is formed with a relatively solid inner anvil portion with an annular diaphragm-like outer skirt connected and extending outward therefrom, the outer skirt positioned to contact the bulkhead before the anvil portion, and the processor is calibrated to generate a first closed sound when the outer skirt contacts the bulkhead and a different after touch sound when the anvil portion contacts the bulkhead.

19. The hi-hat cymbal of claim 11, further including a spring positioned between the stationary

housing and movable housing, and the processor is calibrated to generate a first closed sound when the movable housing first contacts the spring and a different after touch sound when the movable housing compresses the spring against the stationary housing.

20. The hi-hat cymbal of claim 11, further including at least one additional electronic sensor selected from the group consisting of a piezo-electric sensor and a force sensing resistor sensor, the at least one additional electronic sensor being mounted to a surface on the stationary housing that the movable housing impacts so as to generate an electronic signal upon impact between the movable housing and the stationary housing.
